11.1.5 Mechanical wave motion and its types

Wave motion can be classified into two types

a. **Mechanical wave** – Waves which require a medium for propagation are known as mechanical waves.

   Examples: sound waves, ripples formed on the surface of water, etc.

b. **Non mechanical wave** – Waves which do not require any medium for propagation are known as non-mechanical waves.

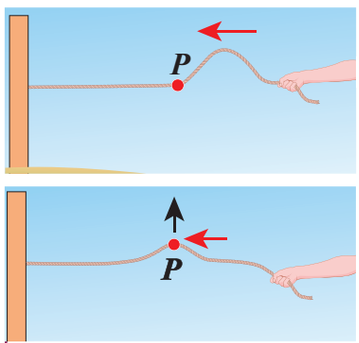
   Example: light waves, Infra red rays etc.

Further, waves can also be classified into two types

   a. Transverse waves

   b. Longitudinal waves

11.1.6 Transverse wave motion



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Figure 11.6 Transverse wave

In transverse wave motion, the constituents of the medium oscillate or vibrate about their mean positions in a direction perpendicular to the direction of propagation (direction of energy transfer) of waves as shown in Figure 11.6.

Example: light (electromagnetic waves)

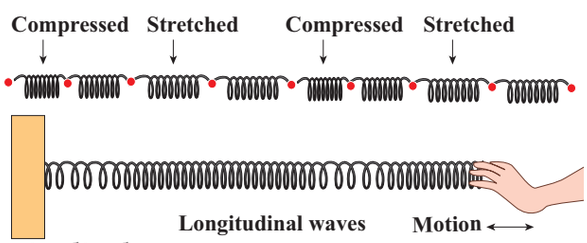
11.1.7 Longitudinal wave motion

In longitudinal wave motion, the constituents of the medium oscillate or vibrate about their mean positions in a direction parallel to the direction of propagation (direction of energy transfer) of waves as shown in Figure 11.7.

Example: Sound waves travelling in air.

Discuss with your Teacher

- Tsunami (pronounced soo-nah-mee in Japanese) means Harbour waves.
- Tsunami is a series of huge and giant waves which come with great speed and huge force. What happened on 26th December2004 in southern part of India? - Discuss
- Gravitational waves and LIGO (Laser lnterferometer Gravitational wave Observatory) experiment.
- Nobel Prize winners in Physics 2017 are Prof. Rainer Weiss, Prof. Barry C. Barish and Prof. Kip S. Thorne for decisive contributions to the LIGO detector and observation of gravitational forces.



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Figure 11.7 Longitudinal waves

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